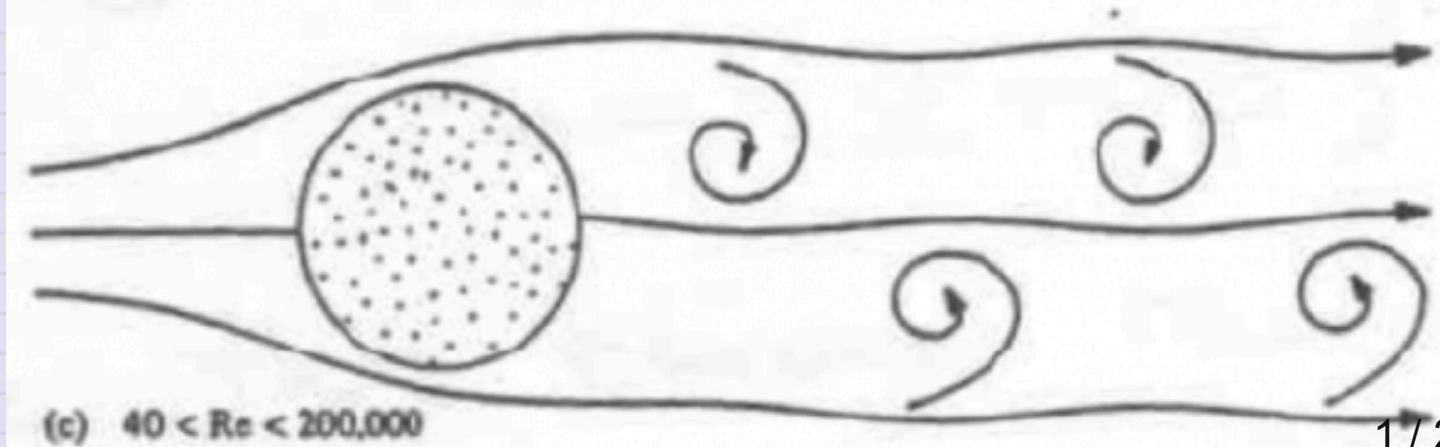
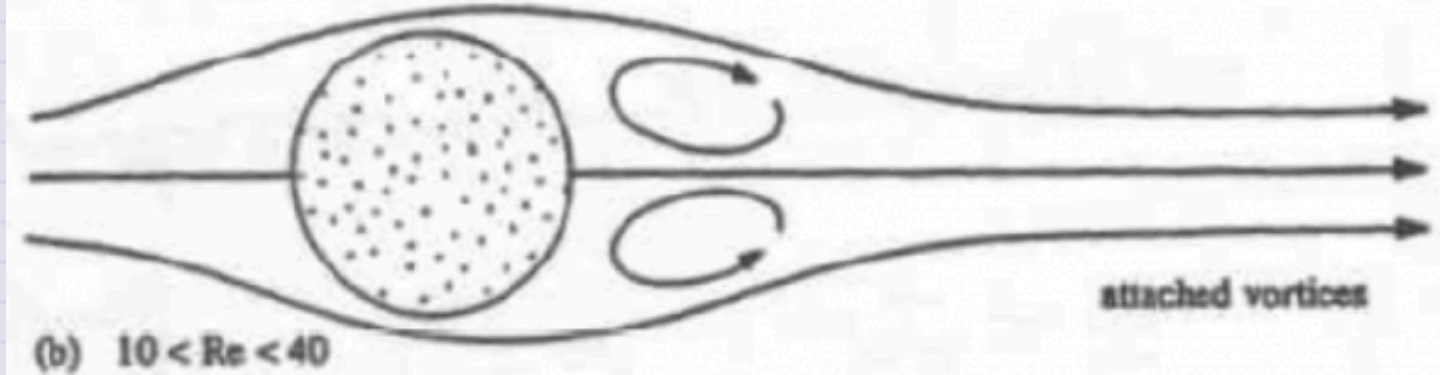
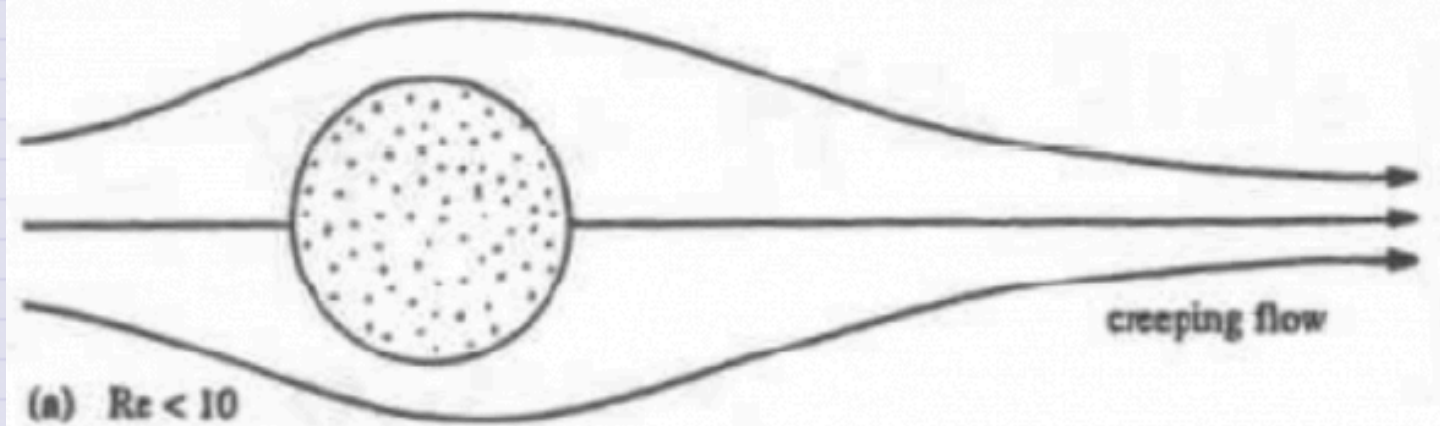


Day 06 - Making Classical Models



Plane Polar Coordinates Warm-Up (5 minutes)

We introduced plane polar coordinates (r, ϕ) . For any position vector, $\vec{R}$, we can write:

$$\vec{R} = |\vec{R}| \hat{r} = r \hat{r}$$

where r is the magnitude of $\vec{R}$, and $\hat{r}$ is the radial unit vector.

Find $\dot{\vec{R}} = \frac{d\vec{R}}{dt}$. Get as far as you can. Our answer will be in terms of $\hat{r}$ and $\hat{\phi}$.

Remember the chain rule and Cartesian unit vectors are fixed in space/time

$$\begin{aligned} \hat{r} &= \cos(\phi) \hat{x} + \sin(\phi) \hat{y} & \hat{\phi} &= -\sin(\phi) \hat{x} + \cos(\phi) \hat{y} \\ \frac{d}{d\phi} \cos \phi &= -\sin \phi & \frac{d}{d\phi} \sin \phi &= \cos \phi \end{aligned}$$

Announcements

- Homework 2 is due Friday
- Homework 3 is now posted
- Office hours, for now (Mihir-MN; Danny-DC):
 - Tuesday 6-8pm (MN, Zoom)
 - Thursday 6-8pm (MN, Zoom)
 - Friday 2-4pm (DC, 1248 BPS)
- Zoom Link: <https://msu.zoom.us/j/96882248075>
 - password: phy321msu

Seminars this week

TUESDAY, September 9, 2025

High Energy Physics Seminar, 1:30 pm, 1400 BPS, Host ~ Joey Huston

Speaker: Joshua Isaacson, MSU

Title: Single pion-production and pion propagation in Achilles

Organized by: Joey Huston, Sophie Berkman and Brenda Wenzlick

Seminars this week

WEDNESDAY, September 10, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Rachael Roettenbacher, University of Michigan

Title: Imaging Spotted Stars for an Improved Understanding of Stars and Exoplanets

Zoom Link: [https://msu.zoom.us/j/887295421?](https://msu.zoom.us/j/887295421?pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09)

[pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09](https://msu.zoom.us/j/887295421?pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09)

Meeting ID: 887-295-421

Passcode: 002454

Seminars this week

WEDNESDAY, September 10, 2025

FRIB Nuclear Science Seminar, 3:30pm., FRIB 1300 Auditorium and online via Zoom

Speaker: Suzanne Lapi of the University of Alabama at Birmingham

Title: Development of new isotopes for theranostic applications

Please see website for full abstract.

Please click the link below to join the webinar:

Join Zoom: [https://msu.zoom.us/j/96485010083?
pwd=O0rXwspn80aYGEl06QEZag6Ao4siq7.1](https://msu.zoom.us/j/96485010083?pwd=O0rXwspn80aYGEl06QEZag6Ao4siq7.1)

Meeting ID: 964 8501 0083

Passcode: 261744

Seminars this week

THURSDAY, September 11, 2025

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: Laura Chomiuk, MSU

Title: Fall 2025 Physics and Astronomy Kickoff

Background:

For more information and to schedule time with the speaker, see the colloquium calendar at <https://pa.msu.edu/news-events-seminars/colloquium-schedule.aspx>

Zoom Link: <https://msu.zoom.us/j/94951062663>

Password: 2002 Or complete link: [https://msu.zoom.us/j/94951062663?](https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1)

[pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1](https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1)

Seminars this week

FRIDAY, September 12, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, In Person

Speaker: Jean Paul Sadia, MSU

Title: Introduction to Quantum Information and Computation

Full Scheule is at: <https://sites.google.com/msu.edu/quic-seminar/>

For more information, reach out to Ryan LaRose

Seminars this week

FRIDAY, September 12, 2025

IReNA Online Seminar, 2:00 pm, via Zoom.

Hosted by: Artemis Tsantiri (University of Regina, Canada)

Speaker: Lorenzo Roberti, INFN-LNS, Italy/Konkoly, Observatory Hungary

Title: Carbon-Oxygen Shell Mergers in Massive Stars

Zoom Link: <https://msu.zoom.us/j/827950260>

Password: JINA

AI Policy Proposals

- **Proposal 1:** We adopt a policy that does not allow AI use at all.
- **Proposal 2:** We adopt a policy that allows AI use for brainstorming, help, and editing.
- **Proposal 3:** We adopt a policy that allows AI for use in nearly any way.
- **Proposal 4:** We adopt a policy that allows AI for use in any way with no documentation required.

Updated AI Policy

We have only 12 votes! Vote now, please.

<https://forms.cloud.microsoft/r/0GT4umz7qY>



Goals for Week 3

- Be able to answer the following questions.
 - What is Mathematical Modeling?
 - What is the process for analyzing these models?
- Be able to solve "Simple" Motion Problems with Newton's Laws.

Modeling Video



Source: <https://www.youtube.com/watch?v=dkTncoPqo5Y>

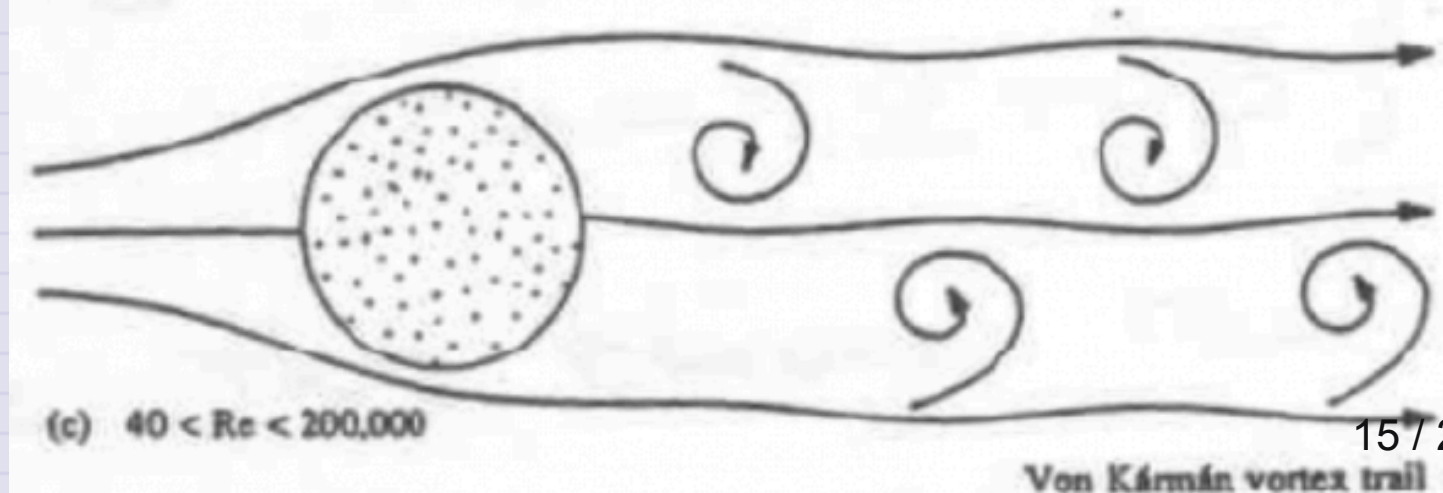
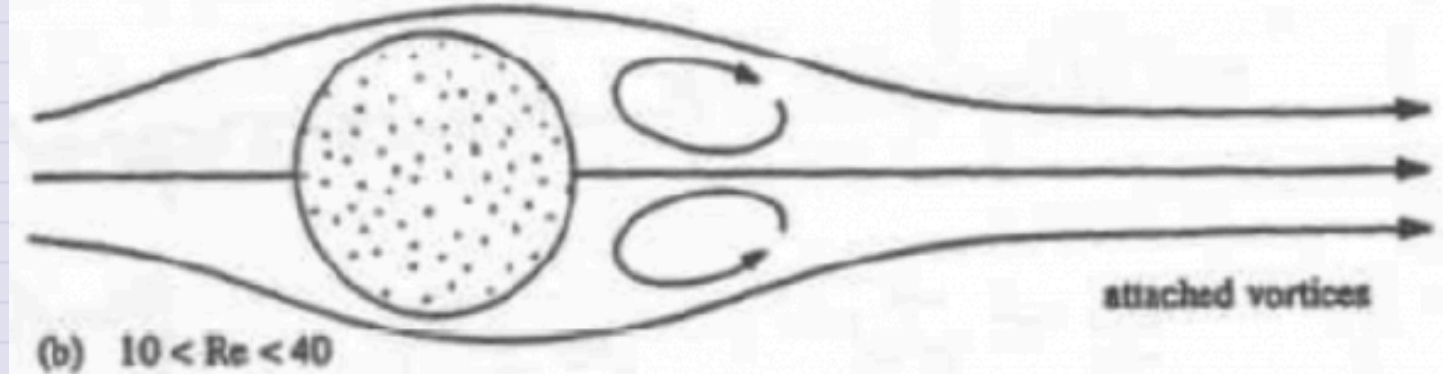
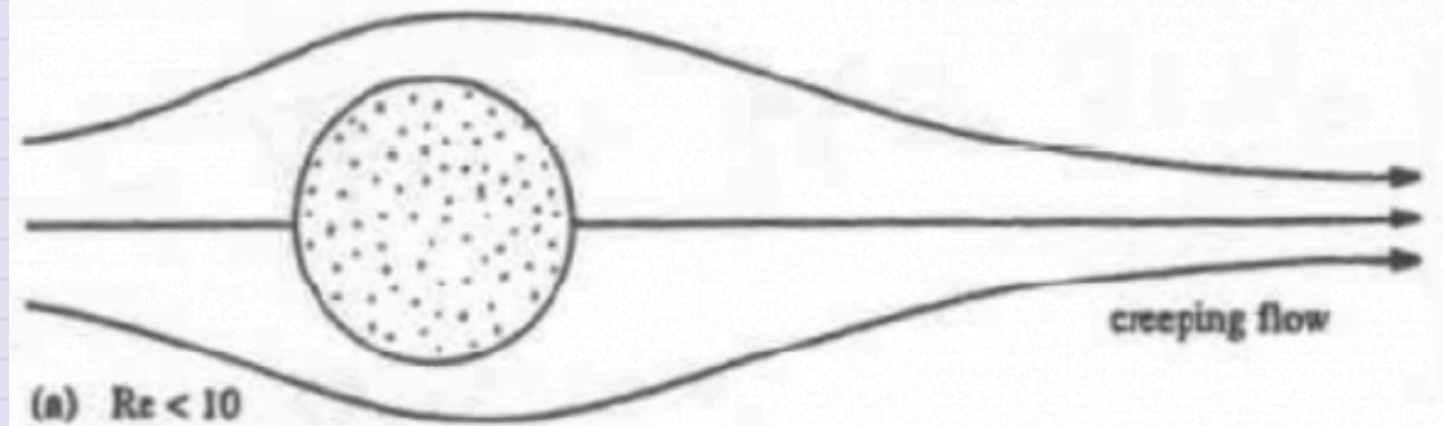
What is your experience with modeling?

Take 2-3 min to think about your prior physics classes

- What models have you used? What makes that a model?
- What made a that model good or not so good?
- What kinds of things could you do to make a better model?

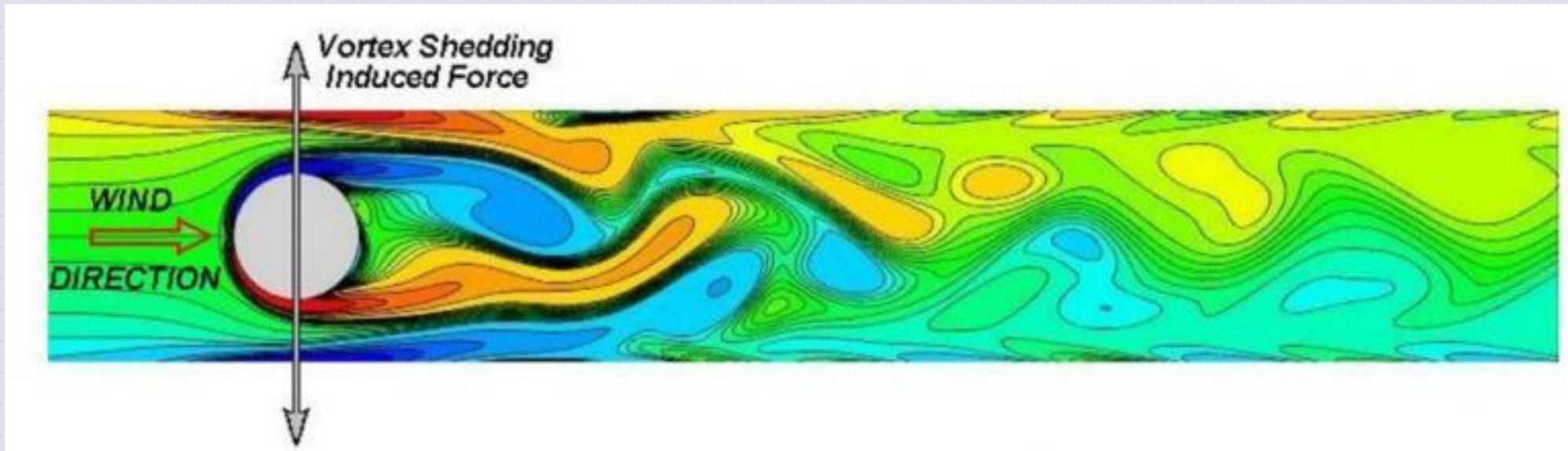
Vortex Shedding

- At higher Reynolds numbers, flow around objects becomes unstable.
- This instability can lead to the formation of vortices.
- This "shedding" of vortices can lead to vibrations and noise.



Model of vortex shedding behind a cylinder

- Controlling vortex shedding is important in many engineering applications.



Giosan, Ioan, and P. Eng. **"Vortex shedding induced loads on free standing structures"**
Structural Vortex Shedding Response Estimation Methodology and Finite Element Simulation 42
(2013).

Renewables: Wind Turbines

Thorntonbank Wind Farm

North Sea off the coast of Belgium

Notice the cylindrical shape of the support structure.



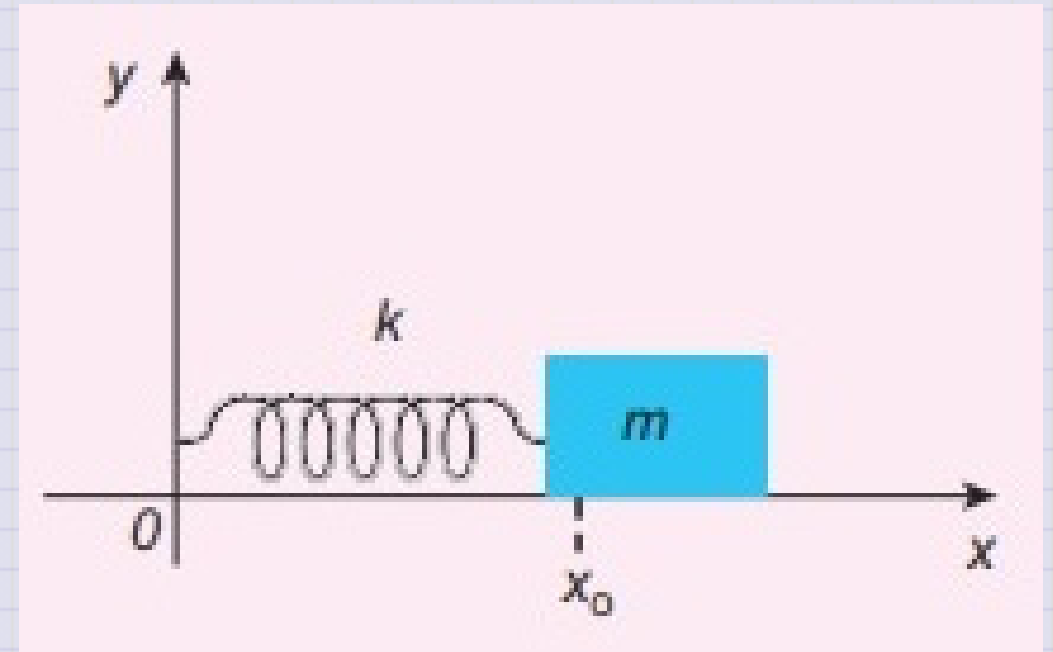
Clicker Question 6-1

The SHO is a useful model:

$$m\ddot{x} = -kx.$$

Assume the **restoring force is anti-symmetric** about the equilibrium position, what is the next term model?

1. $\sim x^2$
2. $\sim x^3$
3. $\sim x^4$
4. $\sim x^5$



Clicker Question 6-2

Assuming a **linear model** for Air Resistance $\sim bv$, we obtained this EOM for a falling ball:

$$\ddot{y} = -g + \frac{b}{m}\dot{y}$$

What happens when $\ddot{y} = 0$?

1. The ball stops moving ($v = 0$).
2. The ball reaches a velocity of mg/b .
3. The ball reaches a terminal velocity.
4. I'm not sure.

Clicker Question 6-3

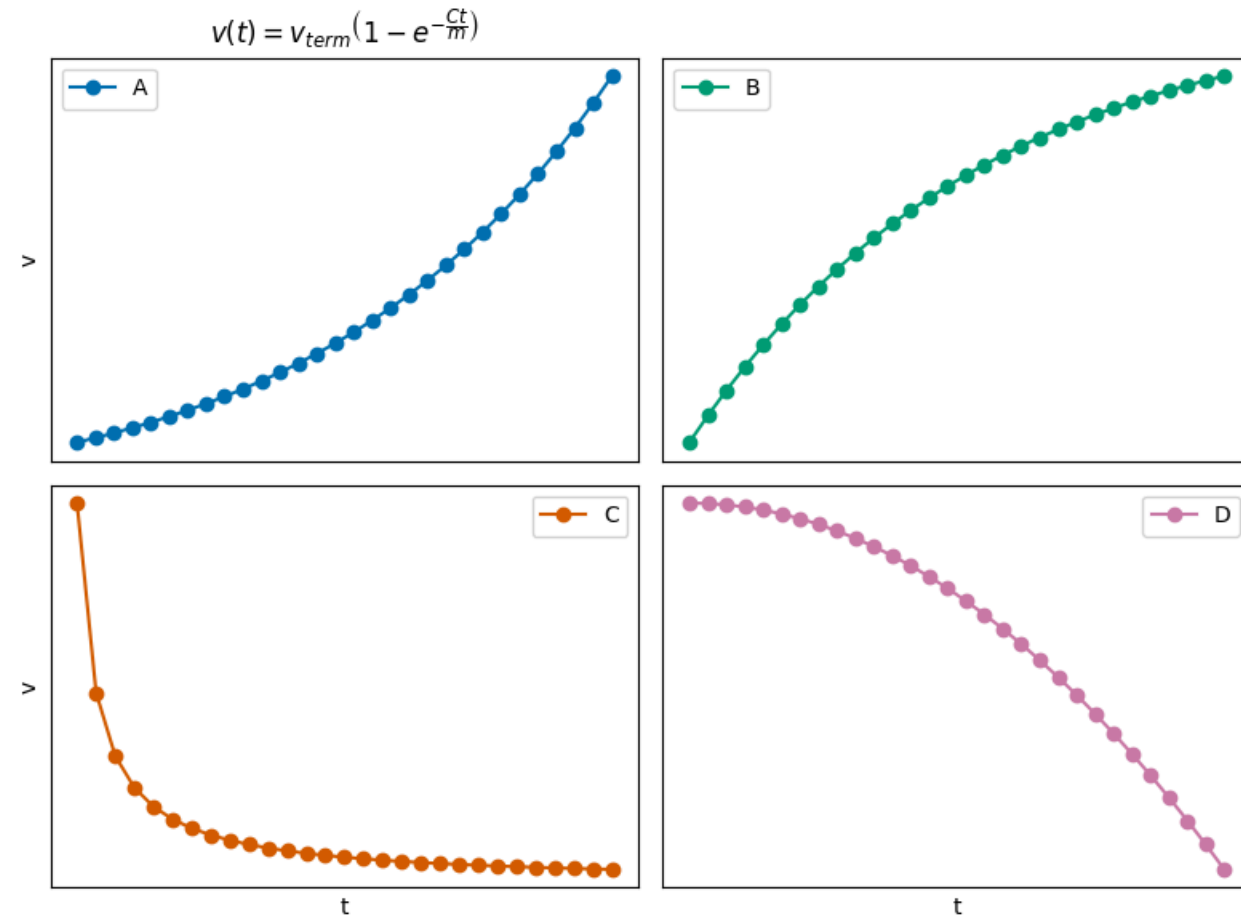
For the system of **Linear Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \left(1 - e^{-\frac{bt}{m}} \right)$$

where $v_{term} = \sqrt{\frac{mg}{b}}$.

CQ 6-3

Which sketch could be correct for the velocity of the ball?



Clicker Question 6-4

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = (mg/c)^{1/2}$. Do the units make sense? What are the units $[gt/v_{term}]$?

1. Yes, the units for $[gt/v_{term}]$ are m/s; v **and** $(mg/c)^{1/2}$ **have the same units.**
2. No, the units for $[gt/v_{term}]$ are m/s; v **and** $(mg/c)^{1/2}$ **have different units.**
3. Yes, the units for $[gt/v_{term}]$ are unit-less; v **and** $(mg/c)^{1/2}$ **have the same units.**
4. No, the units for $[gt/v_{term}]$ are unit-less; v **and** $(mg/c)^{1/2}$ **have the different units.**

Clicker Question 6-5

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/c}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.